



Q1: [20] Three identical charges of charge Q are located at the vertices of an equilateral triangle of side length a . Determine the force on one of the charges due to the other two.

Q2: [40] A total charge of Q_1 is distributed uniformly along a semicircular ring as shown in Figure 1. Two point charges, each of magnitude Q_2 , are situated as shown. The surrounding medium is free space.

- (a) Find Q_2 in terms of Q_1 so that the potential Φ at the center of the ring is zero.
- (b) Find Q_2 in terms of Q_1 so that the electric field $\mathbf{E}$ at the center of the ring is zero.

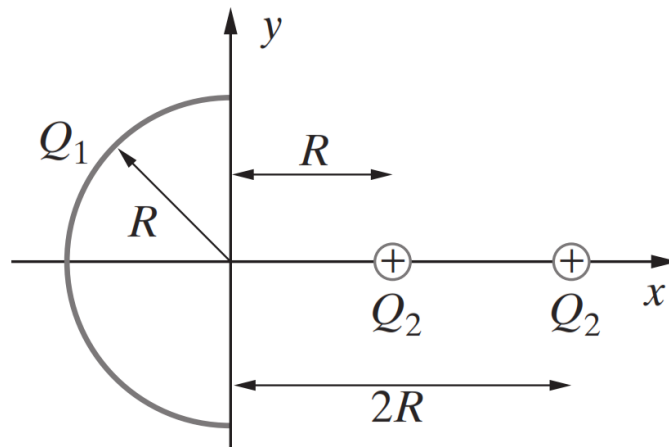


Figure 1: Figure for Question 2.

Q3: [40] The space between two concentric spheres of radii a and b ($a < b$) in free space is charged to a volume charge density given by

$$\rho(r) = \frac{K}{r^2}; \quad a \leq r \leq b$$

where K is a constant. (a) Find the total charge in the shell. (b) Find the electric field at all points in space. (c) Find the electric potential at all points in space. (d) What happens if $b \rightarrow a$?